Table 1: GCB component ablations and pruning comparison on Gwilliams zero-shot. † *w/o norm* R@1 is inflated by sentence-length bias: bucket-size gain correlates with bucket size (Spearman $r{=}0.27$, $p{<}0.001$), inconsistent with the length-leakage-controlled results in Fig. 6. Hard pruning sets non-top-S candidate logits to $-\infty$; GCB-full is the reference.

Method	R@1	R@5	R@10	MRR	MedR
Baseline	0.441	0.657	0.736	0.542	2
Component ablations					
GCB-full	**0.527**	**0.714**	0.777	**0.614**	1
w/o gate	0.378	0.640	0.728	0.498	3
w/o norm†	0.712	0.856	0.885	0.779	1
Pruning comparison ($S{=}3$)					
Hard pruning	0.505	0.623	**0.823**	0.582	1

Table 2: GCB Full vs. Baseline across datasets and backbones. Point estimates are mean query-level metrics across runs. Parentheses show 95% paired cluster-bootstrap CI for the across-run mean GCB improvement, using sentence as the resampling unit within each run; the interval is reported on the absolute GCB metric scale. $^{*}p{<}0.05$ for the across-run mean improvement. R@10 and MedR are point estimates only. Bold: best backbone per metric. zero-shot: 3 seeds; session-isolation: 5 folds $\times$ 3 seeds.

Backbone	Baseline				+GCB Full			
	R@1	MRR	R@10	MedR	R@1	MRR	R@10	MedR
Gwilliams, zero-shot decoding ($N{=}1{,}464$ candidate windows)								
Dense CNN	**0.444**	**0.544**	**0.736**	**2.0**	**0.521 (0.516–0.527)***	**0.608 (0.604–0.613)***	**0.773**	**1.0**
Exp-dilated ($d{=}5$)	0.419	0.519	0.711	**2.0**	0.506 (0.501–0.512)*	0.592 (0.588–0.597)*	0.754	1.3
Exp-dilated ($d{=}10$)	0.396	0.497	0.692	3.0	0.462 (0.457–0.468)*	0.553 (0.549–0.558)*	0.727	2.0
Gwilliams, session-isolation ($N{=}7{,}011$ candidate windows, 5 folds $\times$ 3 seeds)								
Dense CNN	**0.195**	**0.265**	**0.401**	**28.2**	**0.202 (0.200–0.204)***	**0.273 (0.270–0.275)***	**0.408**	**27.1**
Exp-dilated ($d{=}5$)	0.162	0.230	0.361	38.1	0.167 (0.167–0.168)*	0.236 (0.235–0.236)*	0.368	37.0
Exp-dilated ($d{=}10$)	0.151	0.217	0.345	42.9	0.154 (0.154–0.155)*	0.221 (0.221–0.222)*	0.351	42.0
MOUS, zero-shot decoding ($N{=}825$ candidate windows)								
Dense CNN	**0.217**	**0.302**	**0.466**	**13.3**	**0.286 (0.283–0.289)***	**0.370 (0.368–0.373)***	**0.528**	**8.3**
Exp-dilated ($d{=}5$)	0.161	0.240	0.395	21.7	0.215 (0.212–0.218)*	0.293 (0.290–0.295)*	0.440	16.7
Exp-dilated ($d{=}10$)	0.134	0.208	0.351	30.0	0.179 (0.176–0.182)*	0.252 (0.250–0.255)*	0.388	24.7

Table 3: Leakage-control mechanisms in Défossez et al. (2023) and our protocol. The two protocols share the same window formulation and candidate construction; the sole additional control we introduce is an explicit global stimulus-identity split that prevents repeated content from crossing partitions across subjects and sessions.

Mechanism	Défossez et al. (2023)	Our protocol
Window formulation	Fixed 3 s word-anchored windows	Identical
Candidate construction	Audio-window pool de-duplicated by stimulus identity and window index	Identical
Split assignment	Cross-subject/session consistency of repeated stimulus content is not explicitly specified as a global constraint	Global stimulus identity: tuple (audio file, onset, offset), enforced once across all subjects and sessions
Repeated content across subjects / sessions	Not explicitly controlled at the global stimulus-identity level	Prevented by construction: all recordings sharing the same stimulus-identity key are assigned to the same partition, regardless of subject or session

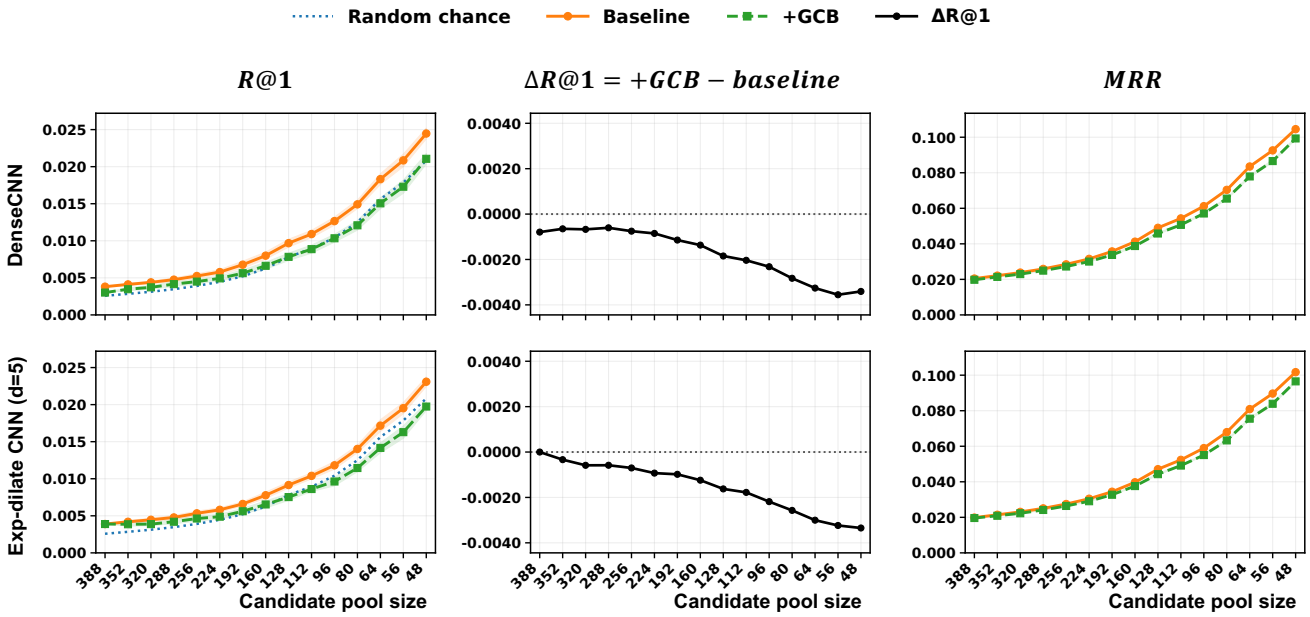

Figure 1: Brennan EEG retrieval across candidate pool sizes. As the candidate pool decreases, absolute R@1 and MRR improve for both backbones. However, GCB remains below the corresponding baseline and ΔR@1 stays negative across all pool sizes.

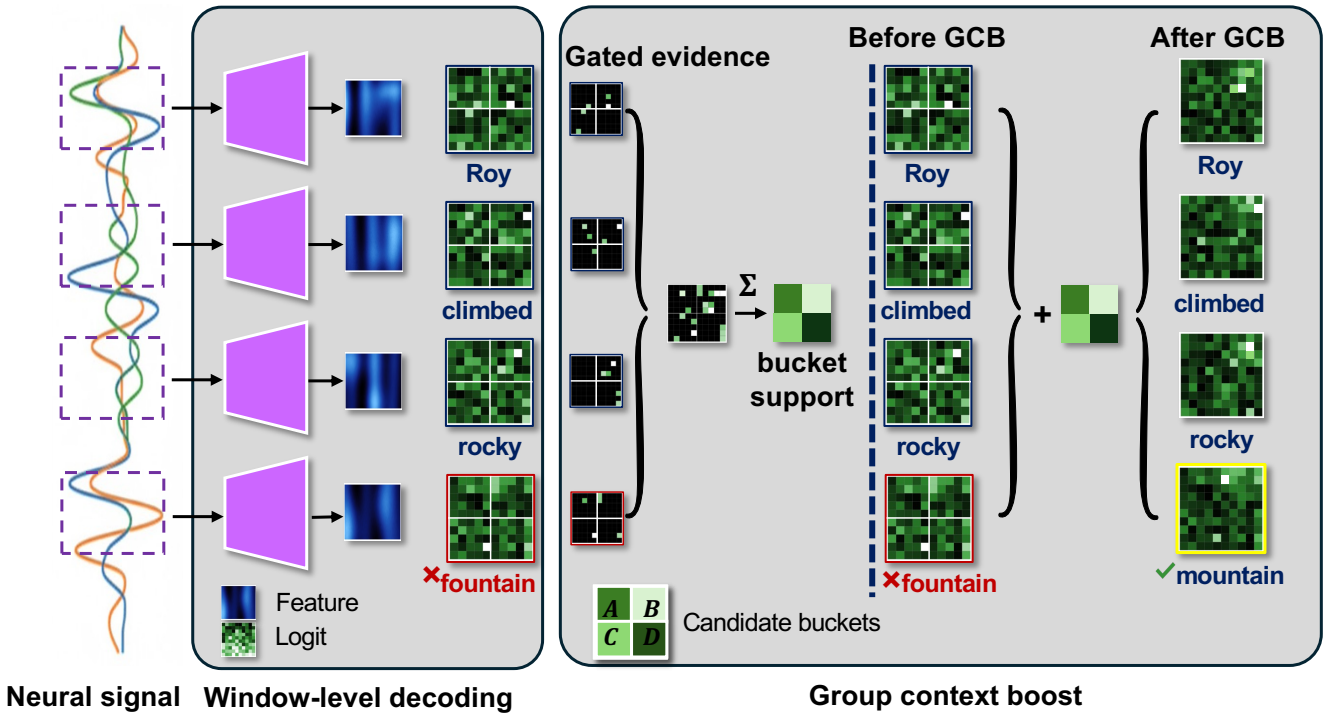

Figure 2: Simplified schematic of the GCB aggregation mechanism. Each neural window is encoded into logits over the candidate pool. GCB retains high-confidence evidence by quantile gating, pools the gated evidence across windows via Σ to compute a shared bucket support, and adds a bucket-wise bias to the original logits. In this example, the fourth window incorrectly ranks *fountain* first before GCB, but *mountain* becomes top-ranked after the shared bucket support is applied.